IS630 Statistical Thinking in Data Science
Project Requirements

1. Overview

The group project challenges you to define a real-world problem and answer analytical questions by applying frequentist and/or Bayesian statistical techniques using Python. Students are expected to apply statistical analysis and visualisation techniques to analyse the data and present a practical and convincing solution to a chosen real-world problem.

Students are free to choose any data set that they are interested to work on but **must position themselves in the perspective of one of the stakeholders who may need to use that data set** (e.g., an organisation, a government body, a customer, etc.). The proposed project should thus perform analytical tasks based on the perspective they have taken on.

The project accounts for 30% of your final grade.

Key deadlines: Please refer to the given schedule during the first lesson.

2. Requirements

Section 0: Title of your Project
A meaningful, relevant and catchy project title.

Section 1: Abstract / Introduction
This is a concise introduction of your project. The introduction should also **include the perspective in which the group has taken on** (e.g., an organisation, a government body, a customer, etc.). This section is usually approximately 150 to 250 words.

Section 2: Problem, Motivation and Objectives
Provide the motivation, objectives, analytical questions, and the impact you aim to achieve.

a. Problem (Motivation, and Objectives)
   i. Clearly define the problem your project addresses.
   ii. Explain the motivation for tackling this problem (the broader significance — why it matters in general, or why it interests you).
   iii. State the main objectives of your project.
b. Analytical Questions
   i. List the analytical questions that guide your project.
   ii. Explain why answering these questions is important for addressing the problem in the context of your target audience.
c. Context and Novelty
   i. Briefly review relevant studies, research papers, articles, or reports, citing them as appropriate.
   ii. Outline your proposed solution and explain how it builds on or differs from previous work (keep it brief for the proposal; provide more detail in the final report or Medium article).
   iii. Explain why your problem or solution is relevant or novel — e.g., its timeliness, practical significance, or the new perspective your team offers.

Section 3: Data Sources
a. Dataset and Variables
   i. Identify and describe your dataset(s), including the source (e.g., public repository, API, company data) and size (rows, columns).
   ii. Provide a simple data dictionary: list the key variables you will use (proposal) or have used (final report), their data types (numeric, categorical, text, datetime), and a short description of what they represent. If the dataset is large, you may describe only the relevant subset.
b. Data Quality and Preparation

i. Briefly describe any data quality issues (e.g., missing values, duplicates, inconsistencies, outliers, biases).
ii. Explain how you will (or did) clean, transform, or manipulate the data to make it suitable for analysis.
iii. If multiple datasets are used, explain how they are combined or how they complement each other (e.g., merged, joined, or used for different purposes).

Section 4: Methodology and Results

a. Approach / Methodology
   i. Describe the statistical and analytical techniques you plan to use (for the proposal) or have used (for the final report or Medium article).
   ii. Provide technical details (e.g., model assumptions, sampling methods, parameter choices, hypothesis tests, and prior specification for Bayesian approaches).
   iii. Justify why these techniques are appropriate for your data and analytical questions.
   iv. Note: In most cases, your project is likely to include a combination of techniques from one or both frequentist and Bayesian philosophies.
b. Results
   i. Present the results of your analysis (e.g., coefficient estimates from regression models, results of hypothesis tests, posterior credible intervals, posterior probabilities).
   ii. Use tables, charts, or other visuals to summarize the key findings clearly.
c. Discussion / Insights
   i. Interpret your results in the context of your problem.
   ii. Explain what the results mean, why they matter, how they answer your analytical questions, and what practical implications they may have (e.g., for decision-making, policy, or future analysis).

Section 5: Conclusion / Summary
Summarise your work, including key insights, recommendations, limitations and future work.

3. Proposal (5%)

The proposal should be a document (**≤ 4 pages**, excluding the cover page (if any) and references) comprising the content related to **Sections #0, #1, #2, #3(a) and #4(a)(i)** of the Requirements.

Submission guidelines: A soft copy proposal (in MS Word or pdf) is to be submitted via eLearn. Only one member from each group needs to submit.

Notes:

- Keep it simple and concise.
- The title can appear at the top of the first page if no cover page is used; no fancy cover page is required.
- Names of team members are not required since eLearn groups already contain your names.
- Submission Guidelines – Proposal.
- A late penalty may be applied, at the instructor's discretion.

4. Project Presentation (10% Group + 5% Individual)

Your team will present in **Session 10**. Each team will have **up to 15 minutes** for the presentation, followed by a **Q&A session** (format determined by your instructor).

- Your presentation should use **up to 17 slides** (1 title/hook slide + 15 content slides + 1 contribution slide showing the contributions of each member). You are not required to present the contribution slide, but it will be used for grading.
- The slides should cover:
  - The **title** of your project
  - **All five sections** of the Requirements
- Provide clear insights and sufficient detail to effectively address the analytical question(s) related to your real-world problem.

- **Submission guidelines:** A soft copy of your slides (in MS PowerPoint or PDF format only) is to be submitted via eLearn. Only one member from each group needs to submit. A late penalty may be applied, at the instructor's discretion.

Notes:

- You may structure your presentation in a way that best fits the unique requirements of your business problem.
- Every team member must present a meaningful portion of the presentation.
- Strive for balance: your presentation should deliver clear content while also being captivating, innovative, and engaging. Focus on condensing key messages so the audience can follow easily, rather than rushing through too many details.
- Manage your time well so that all key sections are covered within the time given.
- Do not read directly from scripts, notes, or mobile devices. This is to prepare you for real-world presentation standards. If needed, you may be asked to put notes down or walk away from notes.
- All team members are expected to understand the entire project. Questions may be directed to anyone, regardless of the section they presented.
- If a team member is absent for a valid reason, another member may present on their behalf. Begin the segment by stating, *"I am presenting on behalf of [name]."* This will not affect the score of the member who helped; the absent member will be given an alternative assessment related to the project.

5. Final Deliverable - Report / Medium Article and Code Base (10%)

5.1 Report / Medium Article

You have the option of submitting a typical report or Medium article (potentially published under MITB's Medium publication – "MITB For All").

Report	Medium Article
[Pros] Everyone knows how to do this (I suppose).	[Pros] Online so anyone around the world can read it. Medium has high domain authority, and your article may appear on Google search. We have articles in MITB's Medium publication (MITB For All) reaching thousands of views. This can be a **portfolio-ready** article to add to your CV.
[Cons] Run-of-the-mill (it will be just another report).	[Cons] Not suitable if using privately-owned data from an organization. Not guaranteed to be accepted for publication in MITB For All (though you may still post it on your own personal profile).
Up to 10 pages (A4) excluding references. Font size 11.	Up to 15 minutes read time, as computed by Medium. This is equivalent to approximately 10 pages.
Use your creativity on the format. If your project is research related, format using LaTeX on Overleaf; IEEE or ACM single-column conference style is recommended.	Target audience is the educated public who may be interested in the domain. The title of your article and first two paragraphs are vital. You need to make the reader see that they benefit from spending the few minutes on the article.
References at the end of the report, properly cited using APA, Chicago, IEEE or ACM (with Author-Year format).	References can either be at the end of the article (not counted towards read time) or directly hyperlinked to the source.
Additional disclosure of Generative AI content can be placed after the References section (this will not count towards the page limit). Use a section titled "**Declaration**" to include statements that are not suitable to be cited formally in the report.	Provide a **separate document** called **"Declaration"** to **disclose** the use of Generative AI content. This document should be uploaded to the eLearn submission DropBox as part of the report submission.

Submission guidelines: A soft copy proposal (in MS Word or pdf) is to be submitted via eLearn. A late penalty may be applied, at the instructor's discretion.	**Submission guidelines:** A hyperlink to the Medium Article is to be submitted via eLearn. The last edit date must not be later than the deadline to avoid late penalty. For submission details, refer to appendix B. A late penalty may be applied, at the instructor's discretion.

Notes:

- **Appendices:** These are entirely optional and will not be explicitly graded. If included (though generally not recommended), they will serve only as supplementary material for the instructor's reference.
- **References:** References form part of the grading criteria. Evaluation will consider the quality, accuracy, and consistency of citations, as well as adherence to ethical and responsible use of all external sources — including materials generated or supported by GenAI tools. Proper acknowledgment of such tools is expected where applicable.
- **Academic integrity:** Students must uphold academic integrity, ensuring the work is original and that all external sources (including GenAI assistance, if used) are appropriately acknowledged. Final deliverables must be completed in line with academic integrity standards and comply with the AI usage guidelines in Section 7 below. Any violation may result in disciplinary action and affect the team's grade.

5.2 Code Base

Please submit all relevant files used in your project, such as the code base, datasets (especially those not accessible online), and any additional files (e.g., Word or Excel sheets used for computations outside of Python). Ensure that all files are clearly labelled (e.g., 01_codes.ipynb, 02_dataset.csv). These artefacts can be compressed into a zip file and submitted via eLearn.

6. Project Grading Rubric (for all components)

You will be assessed based on the criteria below:

Assessment components may encompass (but are not limited to):

- Completeness of solution
- Understanding and application of statistical concepts in the context of the project
- Innovative use of data
- Quality of the evidence-based analysis, insights and recommendations
- Effectiveness presentation delivery
- Proficiency of written communication in the proposal and final report / Medium article
- **Peer evaluation** (Peer evaluation result may affect the whole project, not just a small percentage)

Tips to do well:

This course values your understanding of data and statistical thinking, not the ability to generate surface-level answers.

- Go beyond simple descriptions; ask deeper questions of your data.
- If you find an insight, explore related events or causes.
- If results seem counter-intuitive, investigate why they might be true.
- Always ask "So what?" — why does this matter to the company, customers, or context?
- Develop your higher order thinking skills through critical reflection and explanation.

7. Use of Generative AI in This Project

This assessment allows you to use Generative AI (GenAI) tools such as (but not restricted to) OpenAI ChatGPT, Google Gemini, Microsoft Copilot, Claude, and Perplexity in **limited** and **responsible** ways. Your usage of any such tools must be declared clearly.

GenAI can support your learning but cannot replace your own reasoning and analysis.

7.1 Permit, Permit with Disclosure/Citation or Prohibit

Please read this section carefully.

Task	**Permit without Disclosure or Citation within your deliverables**	**Permit with Disclosure or Citation within your deliverables**	**Prohibit**
General Usage	- Validating of your team's understanding of concepts that are taught in class - Understanding lab solutions to apply to project - Debugging of code - Editing of grammar for clarity	- Brainstorming/ideation - Writing or refining code - Outline the structure of report/presentation - Explaining additional statistical tools or models which are beyond class materials。 - Drafting the first cut of proposal, report, Medium Article or presentation.	- Final editing of the proposal, slides, and report or Medium article - Framing the final analytical questions - Interpretating the analytical results - Evaluating the models' performances - Making recommendations to the audience.
Literature Review	- Nil -	Use reliable AI research tools (all via SMU Library, see **Appendix A for instructions**): - **Research Assistant on SMU Library Search bar** **- Scopus AI** (subscribed by SMU Library), **- Scite** (subscribed by SMU Library), **- SciSpace** (free account that you have to register yourself) Only these Research AI tools are **allowed**, and you **must cite all articles referenced.**	Using general-purpose GenAI (e.g., ChatGPT, Claude, Gemini) for literature review due to potential unreliable sources.

7.2 Rules of Use of GenAI

You must:

- Use GenAI in **ethical** and **responsible** manner as per the University's Code of Academic Integrity
- Critically **evaluate** all GenAI outputs
- **Question** the **accuracy** and **validity** of information and sources
- **Declare** and/or **cite** articles or uses of GenAI in your submission except for the items indicated in the "*Permit without Disclosure or Citation within your deliverables*" column above. One neat way is to keep all your project chats in a single link e.g., https://chat.openai.com/share/examplelink. You can reference the link in the artefacts.

Example of Citation of Use of GenAI (ACM format):

- Within your report: "As suggested in a ChatGPT conversation [15], … [insert information here] …."

- If you are writing a Medium article: “In an exchange with ChatGPT, it was suggested that …” (tweak the tone to suit a general audience).

In the Reference list entry: *[15] OpenAI. 2025. ChatGPT conversation with [Student’s Name]. August 18, 2025. Retrieved from https://chat.openai.com/share/examplelink*

7.3 Risks of Overusing GenAI

Over-reliance on GenAI can **backfire** and is often easy to detect by **plagiarism tools or experienced instructors**. Common issues include (but not restricted to):

- Biased or generic responses
- Shallow application of concepts
- Hallucinated or unreliable information
- Outdated technologies
- Repetition or lack of originality
- Overly formal or uncommon vocabulary that is rarely used in practice

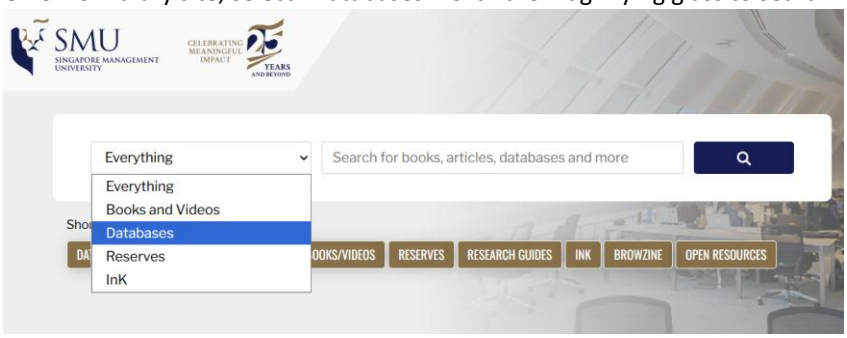
7.4 Academic Integrity

If unauthorised or undisclosed use of GenAI is suspected, your work may be investigated for plagiarism. Confirmed cases will be treated as violations of the University’s Code of Academic Integrity, and penalties will apply according to the University’s policies on cheating.

Generative AI is here to support you in your learning; it is not to replace you.

Appendix A: Use of AI Research Tools offered by SMU Library

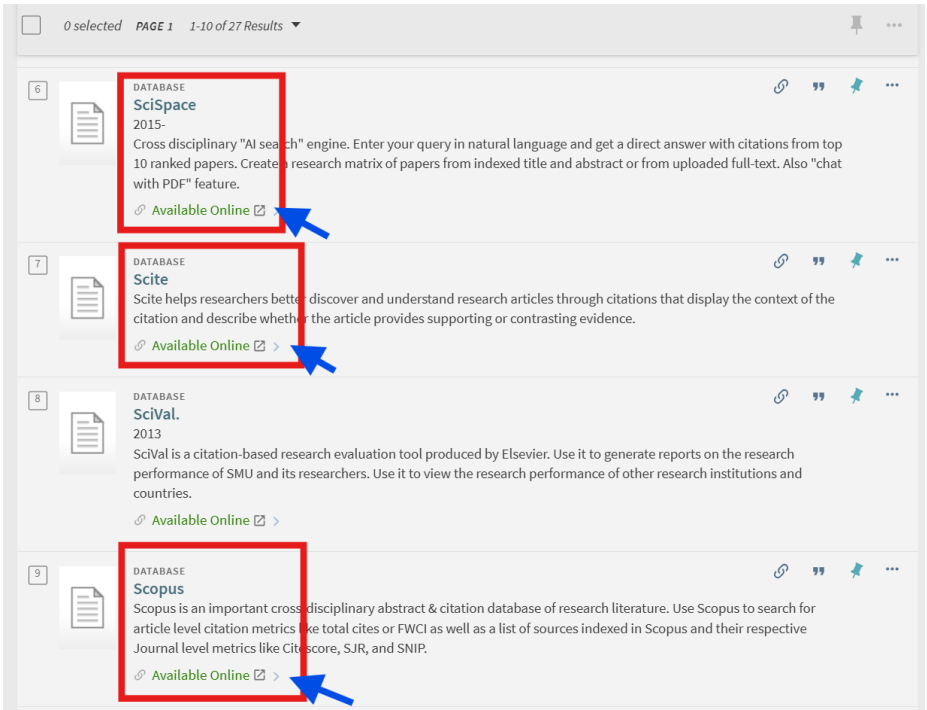
1. On SMU Library site, select “Databases”. Click the magnifying glass to search.



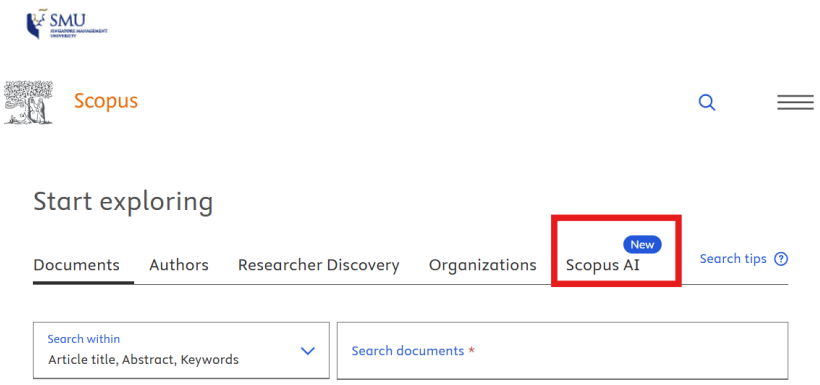
2. To use the Research Assistant, click the tab on the top bar.



3. To use the other Tools, find it in the list within Databases. Example below shows how you may access SciSpace, Scite, and Scopus.

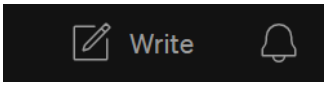


4. To access Scopus AI within Scopus, click the last tab.



Appendix B: Submission process for Medium article

1. Create a free account on Medium.
2. Be added as a writer for the MITB For All publication. Please send one of the editors (jameskohby@smu.edu.sg or kelvinlee@smu.edu.sg) a message from your MITB email, including your medium user id. (Medium only allows one 'official' author to submit, for other contributors, can just include a note within the article itself)
3. To begin writing, click on the 'write' button at the top-right.



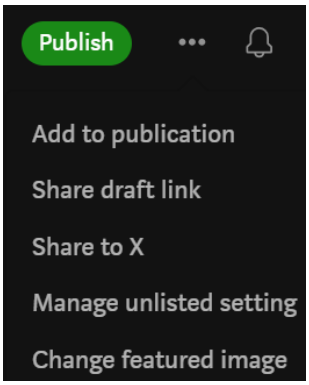
4. The formatting on Medium is straightforward via the UI. To create headings or subheadings, highlight your text and a bar will appear above. Click on the large T for headings, or small T for subheadings. If you want to place a hyperlink, use the third icon (beside the *i*)

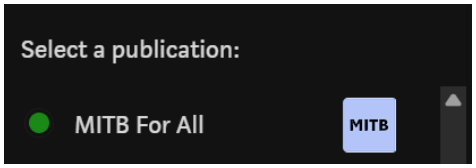


5. To add a code block, click on the plus sign, and select the 5th icon {}.



6. Once done, click on the three dots on the top-right, and select 'Add to publication'. Once you have been added as a writer, you will see an option to select 'MITB For All'.





Fun fact: Well-known international companies such as IBM, Netflix, and Airbnb, as well as Singapore's very own GovTech, and even Barack Obama are on Medium!